

QUANTITATIVE RESEARCH PROJECT

S&P 500 Factor Investing

A Multi-Strategy Research Project

Quality, value, momentum, and trend factors on a frozen S&P 500 universe — from assignment baseline to risk-managed and cost-tested variants.

Common evaluation window: 2016-05-31 → 2026-05-31 (121 months)

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A reproducible research artifact for studying factor-construction and backtest methodology — not an investment recommendation or a production trading system.

Research Question

- Can a transparent, long-only multi-factor S&P 500 strategy outperform two honest benchmarks?
 - (1) random stock selection from the same eligible universe (Monte Carlo null), and
 - (2) the cap-weighted S&P 500 index (^GSPC), on a risk-adjusted basis.
- The discipline: change exactly one design dimension at a time so every performance change has an unambiguous cause.
- Success is judged on Sharpe, drawdown, and statistical separation from random — after realistic research constraints.

Framing

Eight strategy variants share one four-factor backbone. The project is structured to surface honest answers, not optimistic ones: every number is reproducible from frozen raw CSVs, and every deviation from academic best practice is disclosed.

Scope

Academic research artifact — transparent methodology over a single opaque “final” strategy. Not a production trading system.

Research Foundation — Han, Zhou & Zhu (2016)

“A Trend Factor: Any Economic Gains from Using Information over Investment Horizons?” Journal of Financial Economics 122 (2016), 352–375.

- Proposes a trend factor combining short-, intermediate-, and long-horizon price information in one signal.
- Uses normalized moving averages across 11 horizons: 3, 5, 10, 20, 50, 100, 200, 400, 600, 800, 1000 trading days.
- Each month: cross-sectional regression of stock returns on the normalized MA signals; expected returns from the fitted model.
- Forms a long-short factor: long highest expected-return quintile, short lowest (Q5 – Q1).
- In US stocks 1930–2014, the trend factor outperforms standalone short-term reversal, momentum, and long-term reversal — with better crisis behavior than momentum.

Adaptation note

This project does NOT fully replicate the paper's long-short Q5–Q1 factor. It is mostly long-only and S&P 500-focused.

Improved 6 is the closest paper-faithful variant (HZZ cross-sectional trend). Base and Improved 1–5/7/8 use an assignment-driven index-regression trend, not the paper's exact construction.

Data & Universe

Raw file (frozen CSV)	Source	Rows	Span
Daily prices (OHLCV + adj.)	Tiingo	2,292,658	2006–2026
Daily fundamentals (P/E, P/B, PEG)	Tiingo	2,295,967	2006–2026
Statement fundamentals (point-in-time)	Tiingo	3,458,978	2005–2026
S&P 500 constituents	Wikipedia	503	current
Index benchmark ^GSPC	Yahoo	5,030	2006–2026

- 503 current constituents; hard cutoff 2026-05-31.
- Fundamentals joined by date_available — point-in-time, no statement look-ahead.
- Daily → monthly factor panel; ~200 MB via Git LFS.

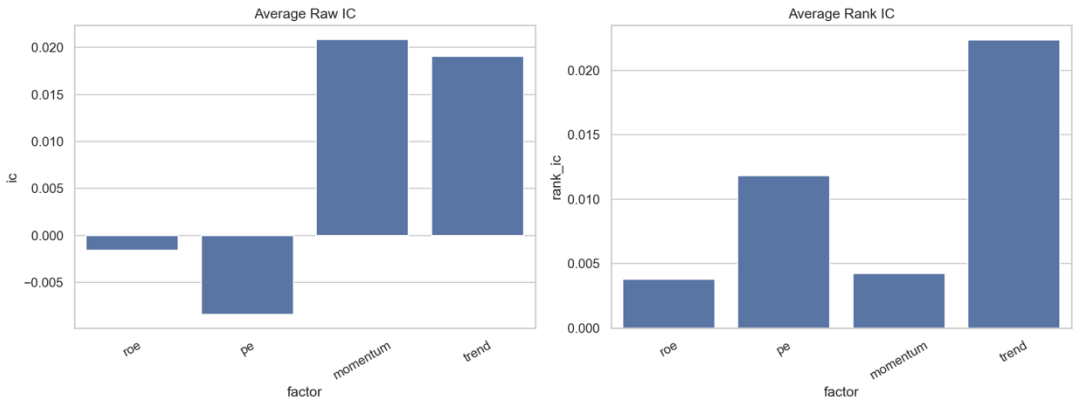
SURVIVORSHIP BIAS: The universe is current constituents only. Delisted/removed names (Lehman, Bear Stearns, GE pre-2018, ...) are absent, and recent additions carry full backfilled histories — survivorship bias inflates Sharpe and depresses drawdown.

Factor Construction

Factor	Family	Definition
ROE	Quality	Point-in-time return on equity
– P/E	Value	Inverted P/E — cheaper scores higher
Momentum	Momentum	12-month price return (t-12 → t)
Trend	Trend	Moving-average trend (HZZ 2016)

- Each factor lagged one month (signal at t → trade at t+1).
- Winsorized at 1st/99th pct within each month, then cross-sectionally z-scored.
- Composite = equal-weighted mean of the four z-scores.
- Three trend implementations — the key methodological fork:
 - base: full-sample index regression — has look-ahead bias.
 - improved 1–5, 7: expanding-window index regression — no look-ahead.
 - improved 6: HZZ stock-level cross-sectional regression — paper-faithful.

Trend is the dominant single factor; quality and value are weak in isolation (Section 9.5).



Information coefficients by factor — trend carries the strongest, most consistent rank IC.

Strategy Specification

Component	Implementation
Universe	503 current S&P 500 securities, frozen through 2026-05-31
Frequency	Monthly signal & rebalance; daily Backtrader execution for stop/take
Factors	ROE, inverse P/E, 12-month momentum, trend
Signal processing	1-month lag, 1st/99th winsorization, cross-sectional z-score
Composite score	Weighted mean of available factor z-scores
Base selection	Top 10 by composite score
Improved 4	Top 10, expanding trend, 5% stop / 30% take
Improved 8	Top 20, 5%-of-equity per position, 5% stop / 30% take
Position sizing	\$100K fixed (base–Imp 7); percent-of-equity (Imp 8)
Execution	Market entries/rebalances; native stop + limit (OCO) exits
Costs	Zero in main runs; vector cost scenarios in Improved 7
Evaluation window	2016-05-31 → 2026-05-31 (121 months)

The Strategy Ladder — Eight Variants

Variant	One change from foundation	Verdict	Message
Base	Assignment full-sample trend	Baseline*	Strong but look-ahead-biased
Improved 1	Expanding no-lookahead trend	Kept	Honest baseline; removes look-ahead
Improved 2	Add 10% stop / 20% take	Kept	Risk-management layer
Improved 3	Dynamic rank-IC weights	Rejected	Adaptive weighting did not help
Improved 4	Walk-forward stop/take 5% / 30%	Winner	Best risk-managed variant
Improved 5	10-mo SMA regime filter	Rejected	Strips useful exposure
Improved 6	HZZ cross-sectional trend	Kept	Paper-faithful; lower per-trade alpha
Improved 7	Time-varying costs (4 & 6)	Study	Cost sensitivity, NOT a strategy
Improved 8	Top-20 + equal-weight 5%	Winner	Wealth-maximizer; lower Sharpe, higher DD

One change at a time. Winners: Improved 4 (risk-managed) & Improved 8 (wealth). *Base look-ahead-biased.

How the Strategy Is Coded

- src/project_core.py — factors, StrategySpec, scoring, vector sim, Backtrader engines, validation, figures.
- src/run_project.py — full pipeline (base → Improved 3).
- run_improved_4_stop_take_sensitivity.py — stop/take grid.
- run_improved_6_hzz_trend.py — HZZ cross-sectional trend.
- run_improved_7_costs.py — cost sensitivity.
- run_improved_8_top_n_sizing.py — equal-weight top-20.

Strategy configuration

```
@dataclass(frozen=True)
class StrategySpec:
    name: str
    weights: dict[str, float]
    top_n: int = 10
    stop_loss: float | None = None
    take_profit: float | None = None
    trend_col: str = "trend_z"
    sizing_method: str = "fixed_cash"
    sizing_target_pct: float = 0.10
```

Backtrader stop / take (OCO)

```
# protective exits, submitted after the entry fills
stop = self.sell(data=d, size=int(sz),
    exectype=bt.Order.Stop,
    price=entry * (1 - stop_loss))
limit = self.sell(data=d, size=int(sz),
    exectype=bt.Order.Limit,
    price=entry * (1 + take_profit),
    oco=stop) # one-cancels-other
```

Improved 8 spec (one config object)

```
core.StrategySpec(
    name=core.IMPROVED_8_STRATEGY_NAME,
    weights={"roe":1,"pe":1,"momentum":1,"trend":1},
    top_n=20, trend_col="trend_expanding_z",
    stop_loss=0.05, take_profit=0.30,
    sizing_method="percent_of_equity",
    sizing_target_pct=0.05)
```


Headline Results

Strategy	BT Sharpe	BT Max DD	BT Final	Vec Final	MC p
Base*	1.36	−6.8%	\$5.78M	\$2.03M	0.013
Improved 1	1.22	−17.2%	\$3.77M	\$3.51M	0.019
Improved 2	1.26	−13.5%	\$2.89M	\$2.72M	0.068
Improved 3	1.21	−15.5%	\$2.68M	\$2.64M	0.157
Improved 4	1.40	−7.7%	\$2.82M	\$2.75M	0.088
Improved 5	1.06	−10.6%	\$2.11M	\$2.15M	0.099
Improved 6	1.02	−11.2%	\$3.16M	\$2.04M	0.290
Improved 8	1.11	−11.9%	\$3.09M	\$3.63M	0.306

Risk-managed

Improved 4 — best Backtrader Sharpe (1.40) and lowest drawdown (−7.7%) among honest variants.

Wealth-oriented

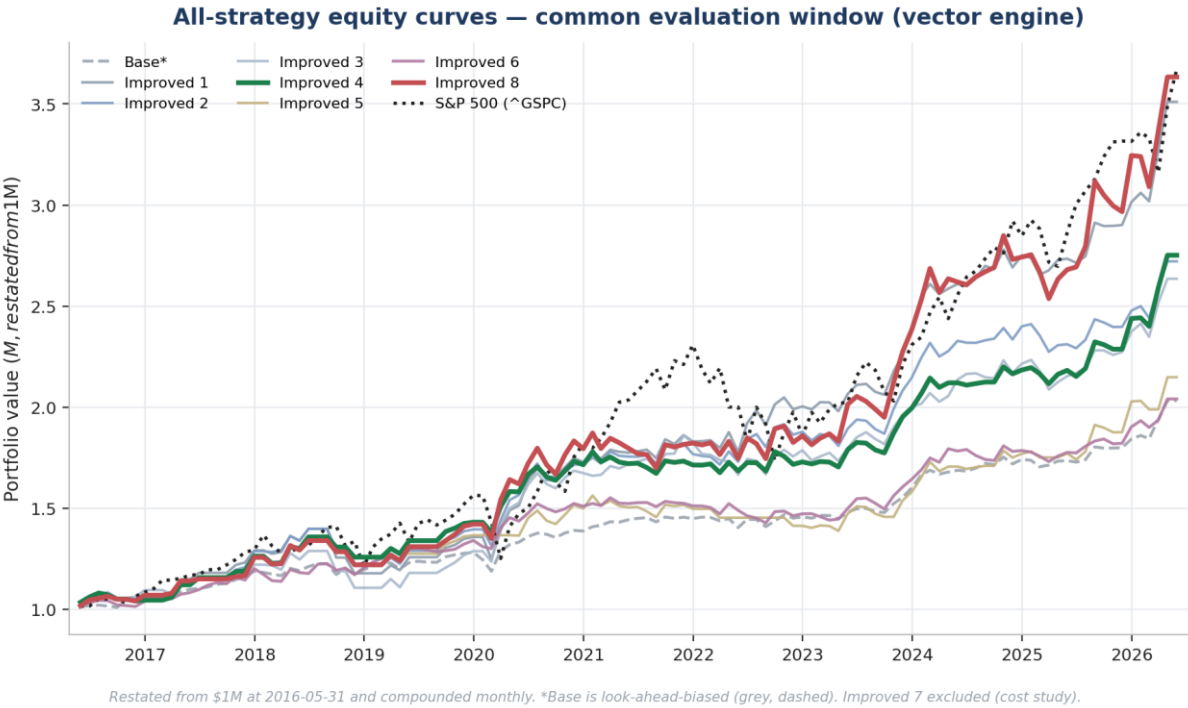
Improved 8 — highest vector final equity (\$3.63M); lower Sharpe because capital is diversified across ~17 names.

Why two final-equity columns?

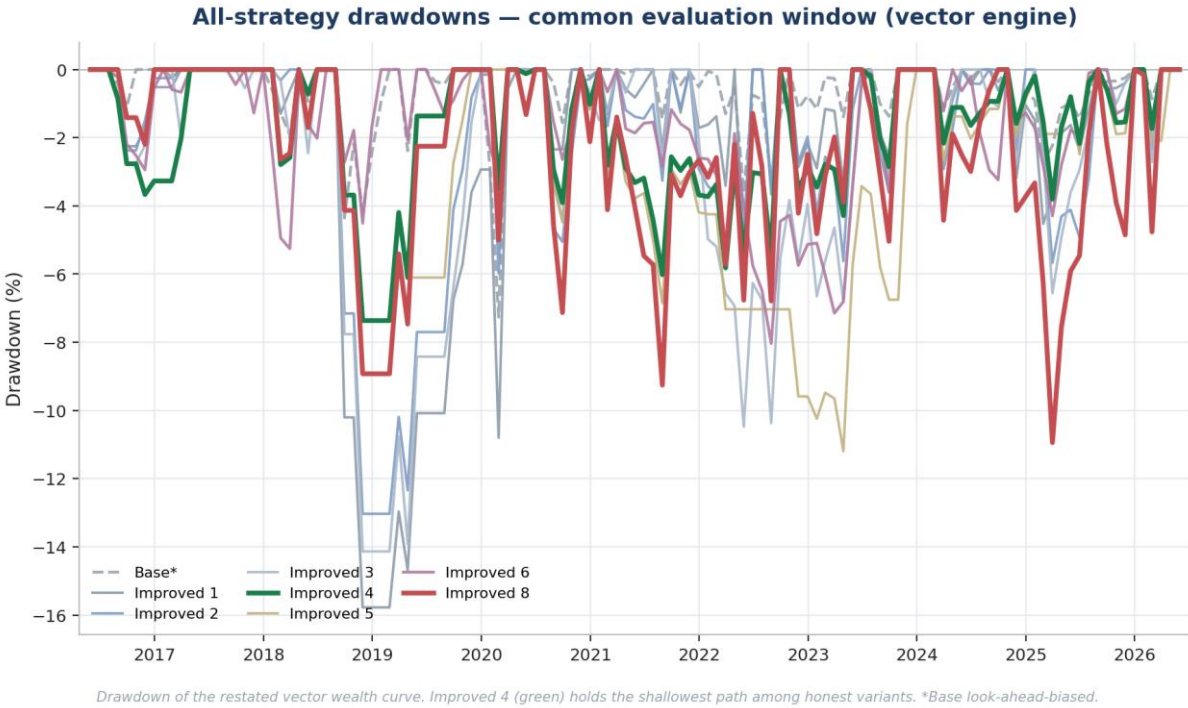
Vector restates wealth from \$1M at eval start; Backtrader tracks absolute value across all years. Sharpes use only the eval window in both — directly comparable.

*Base look-ahead-biased (greyed). Bold = best-in-column among honest variants. Source: README §9.1.

Performance & Risk — All Strategies



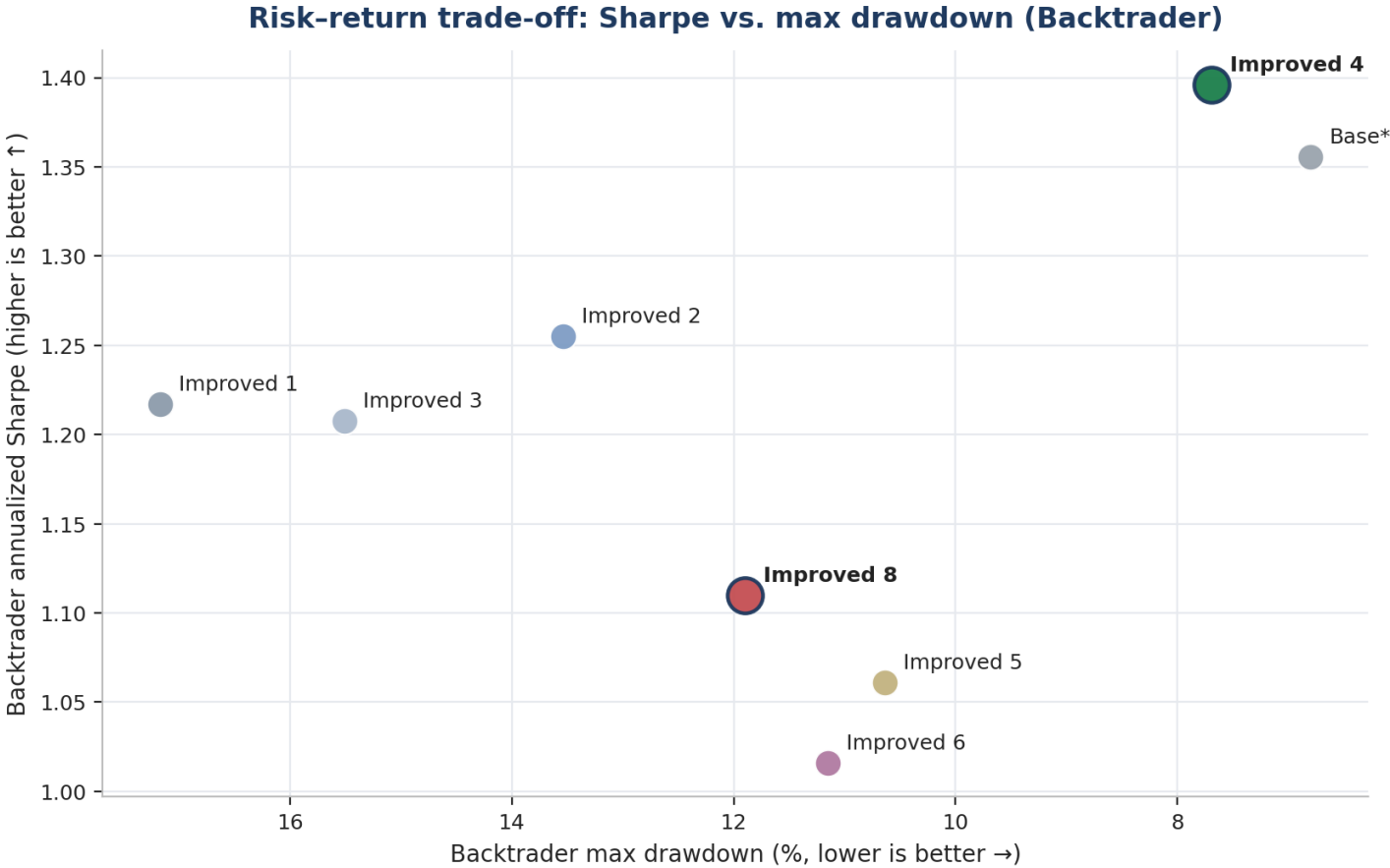
Equity curves vs. ^GSPC (vector, restated from \$1M at eval start)



Drawdown paths (Improved 4 shallowest among honest variants)

All eight variants shown (Improved 7 excluded — cost study). Active portfolios, negative beta to ^GSPC.

Risk–Return Trade-off



Top-right is best (high Sharpe, low drawdown). Improved 4 is the risk-managed leader; Improved 8 trades Sharpe for wealth. *Base look-ahead-biased.

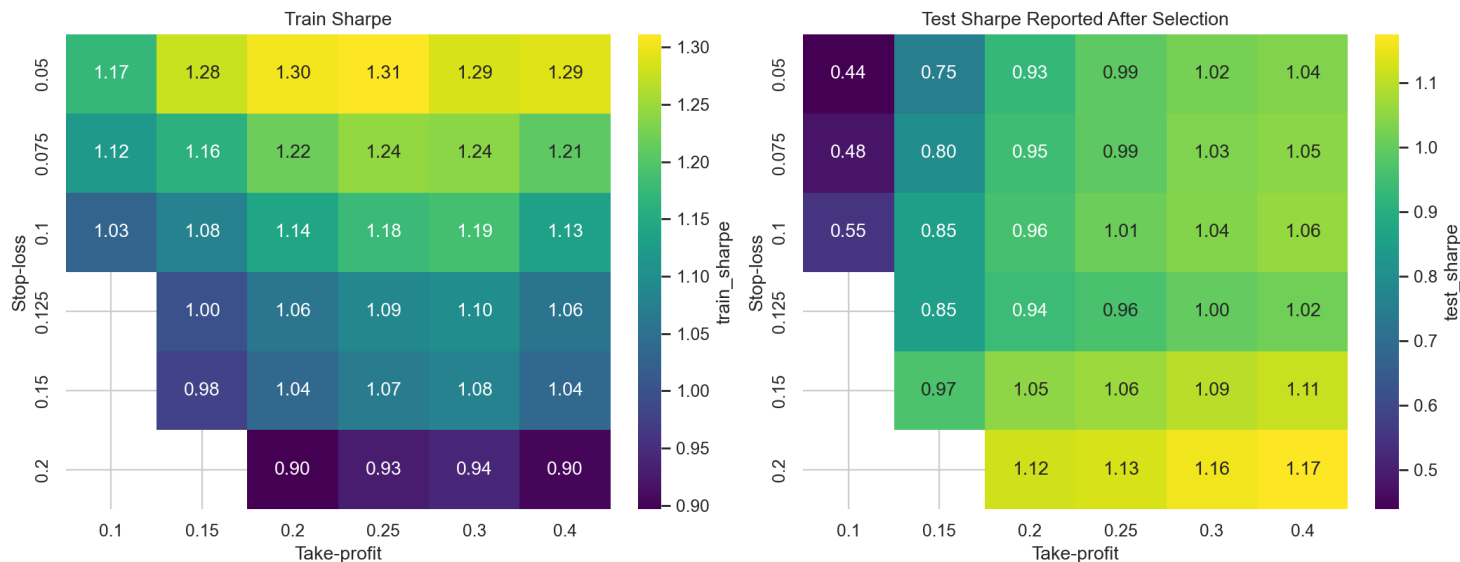
One message

Top-right is best: high Sharpe, shallow drawdown. Improved 4 sits at the efficient corner among honest variants. Improved 8 deliberately moves down-left — it trades risk-adjusted return for absolute wealth.

Caveat

Base appears strong here only because its look-ahead trend suppresses drawdown — not a fair comparison.

Why Improved 4 — Walk-Forward Stop/Take

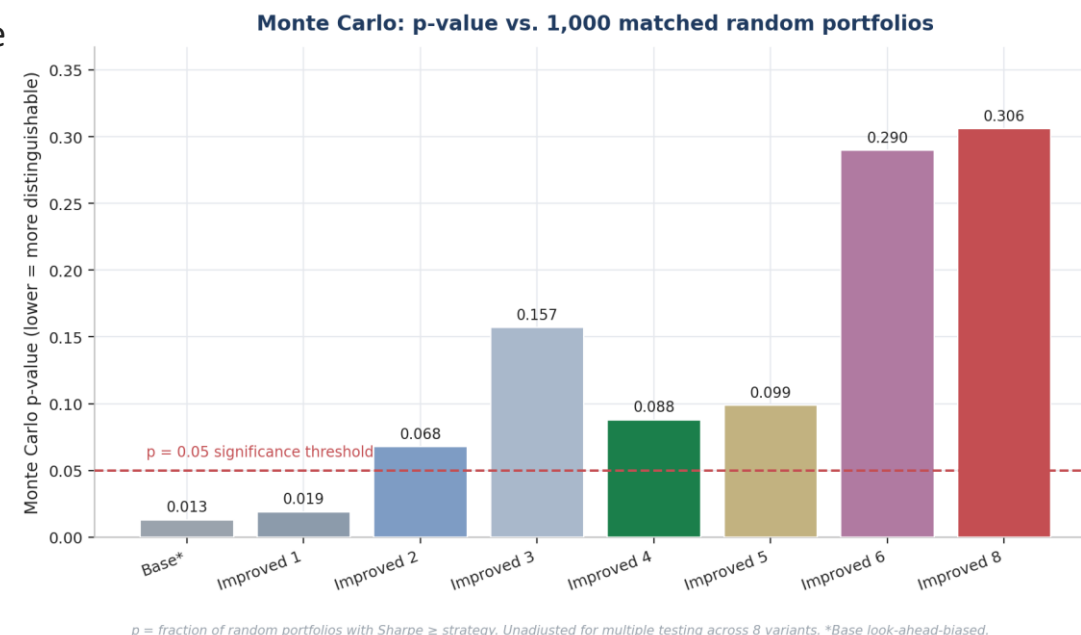


Sharpe across the 6×6 stop/take grid (training data ≤ 2020-12)

- Stop/take chosen by walk-forward on training data through 2020-12, not by full-sample fit.
- Selected pair: 5% stop, 30% take-profit.
- Result: drawdown drops to −7.7% while Sharpe rises to 1.40 — the best risk-managed trade-off in the ladder.
- Honest about tuning: the grid search itself is part of the multiple-testing budget (see Limitations).

Validation Battery

- Monte Carlo — 1,000 random portfolios per strategy from the same eligible universe; p = fraction beating the strategy's Sharpe.
- Block bootstrap — 1,000 paths resampling the strategy's own returns in 6-month blocks.
- Walk-forward — train $\leq$ 2020-12, test 2021+; select on train, report test.
- Benchmark comparison — alpha/beta regression vs ^GSPC.
- Cost sensitivity — time-varying commission + slippage (Improved 7).
- Backtrader — physical broker simulation with native stop/limit (OCO) orders.



Monte Carlo p-value across variants (all 8, base greyed)

HONEST READ: Only base and Improved 1 clear $p < 0.05$ — and p -values are unadjusted for multiple testing. Walk-forward and benchmark-alpha tables cover base and Improved 1–3 only, NOT Improved 4–8.

Validation — Alpha & Walk-Forward

Benchmark alpha vs ^GSPC (eval window)

Strategy	Ann. Alpha	Alpha t	Alpha p	Beta
Improved 1	14.16%	3.96	0.0001	−0.07
Improved 2	11.52%	3.65	0.0003	−0.07
Improved 3	11.16%	3.11	0.0019	−0.07
Base*	7.91%	4.43	0.0000	−0.05

All significant at the 1% level; negative beta = active factor tilt, not index tracking.

Walk-forward (train ≤ 2020-12, test 2021+)

Strategy	Train Shp	Test Shp	Test Cum	Test DD
Improved 2 (selected)	1.14	0.96	58.5%	−5.8%
Base*	1.10	1.30	46.5%	−3.6%
Improved 1	1.03	1.34	103.1%	−5.2%
Improved 3	0.98	0.98	57.4%	−10.5%

Selected on train Sharpe; test reported after selection. A single split is too narrow to crown a winner.

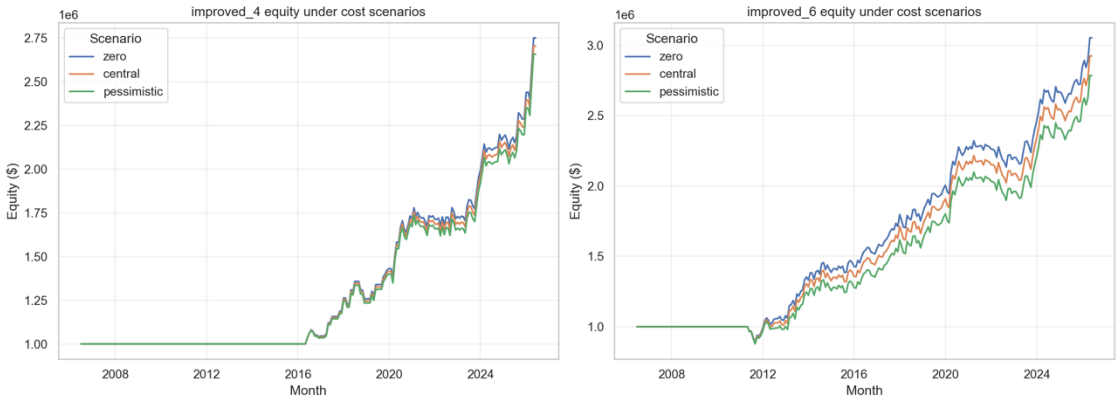
COVERAGE LIMIT: Both tables cover base and Improved 1–3 only — extending alpha and walk-forward to Improved 4–8 is outstanding work.

Sources: results/comparison/strategy_benchmark_comparison.csv, walk_forward_summary.csv

Transaction-Cost Sensitivity (Improved 7)

Strategy (vector Sharpe)	Zero	Central	Pessimistic
Improved 4	1.150	1.118	1.086
Improved 6	1.023	0.985	0.948

- Time-varying per-year schedule (commission + slippage), 2006–2026, from Frazzini-Israel-Moskowitz (2018) and execution research.
- Three scenarios: zero / central / pessimistic (2× central).
- Both finalists survive pessimistic costs; Improved 4 leads at every level. Sharpe drop is small (~0.03–0.07) — modest monthly turnover, tiny size vs S&P 500 ADV.



Equity curves under zero / central / pessimistic costs

SCOPE: This is a vector cost-sensitivity study, not a full dated Backtrader commission/slippage model. Improved 7 is not a standalone strategy.

Source: results/improved_strategy_7/vector_results_grid.csv

Interpretation

Improved 4 — the Sharpe story

Best risk-managed variant. Highest Backtrader Sharpe (1.40) and shallowest drawdown (−7.7%). Concentrated, walk-forward-tuned risk exits.

Improved 8 — the wealth story

Best wealth-oriented sizing variant. Highest vector final equity (\$3.63M) via equal-weight compounding across ~17 names — at the cost of lower Sharpe and deeper drawdown.

- There is no single winner: Improved 4 wins on risk-adjusted return; Improved 8 wins on absolute wealth. The trade-off is exactly what the concentration-vs-diversification literature predicts.
- Trend is the dominant single factor (FMP Sharpe ~0.36–0.39, strongest rank IC); quality and value mainly reduce variance.
- The common-window fix matters: fair Sharpe comparison overturned an earlier spurious “Improved 6 is best” conclusion.
- The defensible claim is relative, not absolute — these strategies beat random and the index on this data, within stated constraints.

Investor Readiness

Real capital today? No — not yet.

Research watchlist / paper portfolio? Yes — Improved 4 and Improved 8.

- Reasons to keep researching: economically sensible signal process; strong historical results on the available data; transparent, multi-lens validation; finalists survive realistic costs.
- Reasons not to allocate yet: survivorship bias; uncorrected multiple testing; execution costs not fully integrated; no sector / liquidity / beta controls; limited walk-forward coverage for Improved 4–8.

Verdict

A cautious investor would not allocate real capital on this deck alone. The work justifies further due diligence, paper trading, and a survivorship-bias-free replication. Improved 4 is the most credible risk-managed candidate; Improved 8 is an interesting wealth-compounding variant — but neither is deployable until the data and validation gaps are closed.

Limitations & Future Work

Limitations

- Survivorship bias — current constituents only; biggest gap.
- Multiple testing — 8 variants, p-values not adjusted (no Hansen SPA / Romano-Wolf).
- Costs — sensitivity study only, not integrated into every backtest.
- Coverage — walk-forward & benchmark alpha cover base + 1–3 only.
- Sizing — fixed-cash gives 4–9 realized names (improves 1–7).
- Production gap — no broker, live data, sector caps, or CI suite.

Future work (priority order)

- Hansen SPA / Romano-Wolf multi-comparison correction.
- Survivorship-bias-free panel via WRDS / CRSP point-in-time membership.
- Long-short paper-faithful HZZ Q5–Q1 portfolio.
- Factor orthogonalization (trend $\perp$ momentum).
- Sector-concentration analysis and caps.
- Top-N sensitivity grid under equal-weight sizing.

Conclusion

A transparent multi-factor S&P 500 ranking process can produce historically attractive returns on the available data. The strongest claim is relative, not absolute: Improved 4 is the best risk-managed variant, Improved 8 the best wealth-oriented sizing variant — and both remain research results until survivorship-free data, multiple-testing correction, and integrated execution costs are added.